The Geometric Engine of Time: The Fokker-Planck Prime Sieve — Companion: Status of Claims, Audit and Open Problems

BioChain AI — SHD-CCP Architecture Research

Consolidated Technical Report (Continued Series)

Abstract

This companion holds everything about the Fokker-Planck prime sieve that is not its statement and proof. The sieve itself, its proofs and the standard it supports are in the release paper (Hoard 2026e). Here are the relation of that paper to its three earlier versions and the sixteen questions those versions raised; the consolidated status of 72 claims of the source note, of the nine companion papers of the SHD-CCP programme and of the release itself; the audit of the source note; and the corrections of the companion papers, with complete proofs where a correction rests on a theorem. Of the companion papers' claims, the golden solitons, the one-sided delta, the 720-step guarantee for arbitrary deltas, the three earlier packet layouts and the South Halo as an integrity check do not survive. The trefoil's D_3 symmetry, periodic closure in the binary polyhedral groups and the coherence score survive in corrected form. The 720-step cycle is examined with the telescoping identity of the cubit installment, which shows that the cyclic holonomy is identically 1 and so cannot test closure. Ten open problems are stated in the manner of a registered report: each with its status, the evidence so far, a test and a decision rule fixed before any new computation or measurement, and a note on whether the release depends on it. It depends on none of them.

Editorial Note. This document is the companion to *The Geometric Engine of Time: The Fokker-Planck Prime Sieve in Quaternionic Lattices* (Hoard 2026e) and is cited in the series as Hoard (2026h). Section and result numbers without a qualifier refer to Hoard (2026e); numbers marked "here" refer to this companion. It follows the conventions of *A Proposal for a Standardization of the Cubit* (Hoard 2026d), including its registered-report format for open claims, and cites the companion papers of the SHD-CCP programme as BioChain AI (2026a–i). All quantitative claims are annotated with status tags: **[PROVEN]**, **[HEURISTIC]**, **[CONJECTURAL]**, **[OPEN]**, **[SUPERSEDED]**, refined by the five further tags of Hoard (2026e), Table 1.

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Epistemic Conventions and Scope

The companion uses the ten tags of Hoard (2026e), Table 1, and its rule for mixed claims: a computed value of a heuristic formula is tagged **[HEURISTIC / COMPUTED]**. A claim of a source keeps the tag that the source's own argument supports, which is not always the tag the source gave itself.

What this companion contains. The history of the installment (Section 1), the status of every claim (Section 2), the audit of the source note (Section 3), the corrections of the companion papers (Section 4) and the open problems (Section 5). It proves results only where a correction rests on them.

What it does not do. It does not restate the sieve; it cites Hoard (2026e) for every positive result. It reviews the companion papers only where the sieve uses them, so a claim not listed here is neither endorsed nor refuted.

Verification. Every number here is reproduced by the script of Hoard (2026e), Appendix A; Section 6 lists the sections that check this companion.

1 Relation to Earlier Versions

The installment reached its release form in four versions. Table 1 lists what each did, Table 2 the questions the earlier versions left open and where the release settles them, and Section

1.3 the corrections made between versions.

1.1 The Versions

Table 1: Versions of the installment.

Version	What it did	Sources added
First edition	Formalized the source note: the lattice master equation, the Gibbs state, absorbing prime traps, the three-square obstruction, the no-collinear-prime-triple theorem, the tier rotation as a Hurwitz unit, and the audit of the source note	The source note; Hoard 2026a
Second edition	Rebuilt the sieve around the companion papers: the Dual-Helix reading, the golden torus and its Fibonacci approximants, the norm-5 hyperbolic tree, the 64-bit packet and the alignment standard	Nine companion papers, BioChain AI 2026a–i; Hoard 2026b, c, d; Butt et al. 2026
Revision	Imported the proofs of the Fifteen Lemmas, adopted the Einstein Tile as the packet, and relabelled the companion papers	The Fifteen Lemmas
Release	Split the installment into the release paper (Hoard 2026e) and this companion, and strengthened four steps with the cubit installment	Further results of Hoard 2026d

Results kept from the first edition are, in release numbering, Definition 2.1, Propositions 2.2 and 2.3, Proposition 4.1, Theorem 4.2, Theorem 8.4 and Proposition 10.1, together with the audit in Section 3 here. The release adds the entropy production and Kolmogorov reading of Section 2.3, Proposition 3.3, the physical scale and precision horizon of Sections 8.1 and 8.2, the medium-free invariants of Section 9.3, and the cube-group statement of Section 9.4.

1.2 Questions of the Earlier Versions

Table 2: Questions raised by the earlier versions and the companion papers, and where they are settled.

#	Question	Raised in	Settled by	Answer, in release numbering	Status
1	Does the lattice geometry compute the primes, or must they be an input?	Source note §5; first edition	BioChain AI 2026a; Hoard 2026a; Fifteen Lemmas, Lemma 14	Nothing needs computing: the sieve proposes and exact tests certify, with a proven wheel bias of 3.75 (Section 3)	**[PROVEN]**; the "computes" claim **[SUPERSEDED]**
2	Three dimensions or four?	Source note §3; first edition	BioChain AI 2026c, 2026d, 2026h; Fifteen Lemmas, Lemma 5	Four: S^3 is parallelizable, the Hurwitz order holds every prime, and the golden-torus approximants give exactly primes that 3D misses (Sections 4, 5)	**[PROVEN]**, **[ESTABLISHED]**
3	How is floating-point drift removed from long runs?	BioChain AI 2026d §5	Fifteen Lemmas, Lemmas 12 and 13	Exact Hurwitz arithmetic has no drift; exact periodic schedules close within 12 repetitions or never (Section 6; Section 4.4 here)	**[PROVEN]**
4	Do golden solitons sit at prime quadrances?	BioChain AI 2026c §4	BioChain AI 2026c	No lattice point lies on the golden torus; exact nodes lie on its Fibonacci approximants, with prime norms 3, 7, 47 and 2207 (Section 5)	**[PROVEN]**; infinitude **[OPEN]**, Problem 1 here

#	Question	Raised in	Settled by	Answer, in release numbering	Status
5	Can the hyperbolic projection run on shifts?	BioChain AI 2026d §8, horizon 2	BioChain AI 2026d; Lubotzky et al. 1988	For the norm-5 generators every step is a permutation, a shift and an add (Section 7.2)	**[PROVEN]**
6	How is a hyperbolic procedural structure generated, addressed and shared exactly?	BioChain AI 2026d §3	BioChain AI 2026d, 2026a; Fifteen Lemmas, Lemmas 8 and 13	Words over 1 ± 2i, 1 ± 2j, 1 ± 2k: free, uniquely factorable, with Ramanujan quotients (Section 7)	**[ESTABLISHED]**, **[COMPUTED]**
7	Does a 64-bit word suffice for the TQF?	First edition	Fifteen Lemmas, Lemma 5 and §6.1; Hoard 2026d, §3.5	Yes on the Form 1 core (36 bits) and on cubit-anchored lattices (53 bits); wide integers only at the $Q \approx 10^{12}$ horizon (Section 8.2)	**[PROVEN]**; horizons **[HEURISTIC / COMPUTED]**
8	Which bit layout is the packet?	BioChain AI 2026e, 2026f, 2026g	Fifteen Lemmas, §6.1	The Einstein Tile (Section 8.1; Section 4.6 here)	**[DEFINITION]**
9	What does TQF pruning remove?	BioChain AI 2026e, Stage 3	BioChain AI 2026e; Hoard 2026a	On a scan line, everything; on (index, value) samples, exactly the zero-second-difference nodes, losslessly (Section 8.3)	**[PROVEN]**
10	How do agents with no shared frame align non-similar data?	BioChain AI 2026e, Stage 1; 2026h	BioChain AI 2026h, 2026a; Fifteen Lemmas, Lemmas 5, 9 and 15; Hoard 2026d; Butt et al. 2026	Invariants non-locally, Horn's quaternion locally, exact certificates under a Merkle root (Section 9)	**[PROVEN]** mathematics; **[HEURISTIC]** application, Problem 8 here
11	Is the coherence Φ a sound integrity score?	BioChain AI 2026h	BioChain AI 2026h, 2026a; Fifteen Lemmas, Theorem 7.5	Frame-invariant but not sign-invariant; Φ_M corrects it; it measures agreement, not truth (Sections 9.3, 9.6)	**[PROVEN]**, **[HEURISTIC]**
12	Does drift along TQF lines sieve?	Source note §4	Fifteen Lemmas, Lemma 7; Hoard 2026d, Proposition 4.8	No: it carries a current and produces entropy while its stationary state stays uniform (Section 2.3)	**[PROVEN]**
13	Is the sieve defined on the infinite Hurwitz lattice?	First edition	Fifteen Lemmas, Lemma 14	Exactly when the Hurwitz partition function converges; for $V = \alpha \ln n$, exactly when $\beta\alpha > 2$ (Section 2.3)	**[PROVEN]**
14	Does a lattice line meet infinitely many prime-norm nodes?	First edition	None	A case of Bunyakovsky's conjecture	**[OPEN]**, Problem 2 here
15	Is the oracle bias above 1 for factoring?	BioChain AI 2026a §5	None	Not settled; the bias proven in Section 3 is for the prime task only	**[CONJECTURAL]**, Problem 3 here

#	Question	Raised in	Settled by	Answer, in release numbering	Status
16	Do trefoil knot invariants predict prime quadrances?	BioChain AI 2026d §8, horizon 1	None	No mechanism found; the trefoil's D_3 symmetry is proven (Section 4.3 here)	**[OPEN]**, Problem 4 here

1.3 Corrections Between Versions

- **Prime coverage of the Form 1 core.** A draft of the revision stated that the Form 1 core reaches all 1,877 primes up to 16,129. The count had included primes with no encodable core. The correct figure is 1,712 of 1,877; the other 165 need a coordinate outside the byte range (Hoard 2026e, Section 8.2; Appendix A, section Y).
- **Citation labels.** The second edition cited the companion papers with Hoard labels. From the revision on they are cited as BioChain AI (2026a–i), as the Fifteen Lemmas cite them, so that Hoard (2026e) can denote this installment.
- **Packet reading.** The second edition read a 32-bit byte core of four signed bytes; the revision replaced it by the Form 1 core of the Einstein Tile. Section R of the verification script keeps the earlier statistics for comparison.
- **Numbering.** In the release, the second edition's Theorem 6.3 and Proposition 6.4 are Theorem 6.1 and Proposition 6.2; its Proposition 6.1 and Theorem 6.2 are Proposition 4.1 and Theorem 4.2 here; its Sections 8.4 and 11 are Sections 4.7 and 2 here, and its Appendix A is divided between Sections 3 and 4 here; and its Appendices B and C are Appendices A and B of Hoard (2026e).

2 Status of Claims

Table 3 records the status of every claim of the source note, of every companion-paper claim the sieve uses, and of the results of the installment, in the series-standard vocabulary. The Source column gives the version in which a result entered: "ed. 1" and "ed. 2" for the first and second editions, "FL" for results imported from the Fifteen Lemmas in the revision, and "release" for results added in the release. Basis numbers refer to Hoard (2026e) unless marked "here".

Table 3: Consolidated status of claims.

Claim	Source	Status	Basis / Note
Lattice Fokker-Planck equation	Source note §1	**[DEFINITION]**	As a master equation; Definition 2.1
Prime potential with minima only at prime Q	Source note §1	**[HEURISTIC]**	The primes enter through V; Section 2.1
Stationary state supported only on prime nodes	Source note §§2, 5	**[SUPERSEDED]**	False for positive diffusion (Proposition 2.2); exact with traps (Proposition 2.3)
Prime basis expansion	Source note §2	**[SUPERSEDED]**	Eigen-expansion of the generator; Section 2.1
Selection rule: $\lambda \to 0$ at primes	Source note §2	**[PROVEN]**	As absorbing traps; Proposition 2.3

Claim	Source	Status	Basis / Note
Inert primes exist but are rare	Source note §3	**[SUPERSEDED]**	None exist; Proposition 4.1
Current flows along TQF lines	Source note §4	**[PROVEN]** as a current; **[SUPERSEDED]** as a sieve	Uniform stationary state; Proposition 2.6
Prime traps on TQF lines, e.g. Q = 2, 3, 5	Source note §4	**[SUPERSEDED]**; norm reading **[OPEN]**	Theorem 8.4; the example is a right triangle (Section 3 here); Problem 2 here
The lattice geometry computes the primes	Source note §5	**[SUPERSEDED]**	Not needed under the Dual-Helix principle; Section 3
$Q \geq 10^{12}$ for 10^{-6} precision	Source note §6	**[HEURISTIC / COMPUTED]**	$\delta\theta \approx 1/\sqrt{Q}$ is the heuristic; Section 8.2
A 64-bit lattice is vastly sufficient	Source note §6	**[PROVEN]** at packet scale and for cubit-anchored lattices; **[SUPERSEDED]** at the horizon	Proposition 8.2; Table 13
Golden metric with shear cφ/N	BioChain AI 2026b §3	**[SUPERSEDED]**	Not modular-invariant (BioChain AI 2026a, §3.2)
Oracle bias β > 1	BioChain AI 2026b §4; 2026a §5	**[PROVEN]** for primes; **[CONJECTURAL]** for factoring	β = 3.75; Proposition 3.2; Problem 3 here
Delta e^{iΔθ₁} e^{jΔθ₂} q stays on the torus	BioChain AI 2026c §3	**[SUPERSEDED]**	Two-sided form, Proposition 5.1; Section 4.1 here
Ergodic covering from $r_1/r_2 = \varphi$	BioChain AI 2026c §2	**[SUPERSEDED]**	Density depends on ω_1/ω_2 (FL Lemma 11); Section 4.1 here
Golden solitons at prime quadrances	BioChain AI 2026c §4	**[SUPERSEDED]**	No lattice point on the torus; Theorems 5.2 and 5.3
Sixteen multiplications by a Strassen scheme	BioChain AI 2026c §5	**[SUPERSEDED]**	Direct product; Proposition 7.3; Section 4.1 here
Quaternion matrices are skew-symmetric	BioChain AI 2026d §2.2	**[SUPERSEDED]**	$L(q)^T L(q) = N(q)I$ (FL Lemma 8); Section 4.2 here
Inverse of every generated Hurwitz path	BioChain AI 2026d §2.1	**[PROVEN]** over $\mathbb{Q}$ only	Inverses of non-units leave the Hurwitz order; Section 4.2 here
SL(2, ℂ) double-covers $SO^+(3, 1)$	BioChain AI 2026d §3	**[ESTABLISHED]**	As the inclusion $SU(2) \subset SL(2, \mathbb{C})$ (FL Lemma 8); Section 4.5 here
D_3 symmetry of the trefoil; one-third storage	BioChain AI 2026d §4	Symmetry **[PROVEN]**; one-sixth tables **[PROVEN]**; data folding **[SUPERSEDED]**	Proposition 4.1 here (FL Lemma 8)
Return to identity at step 720	BioChain AI 2026d §5	**[PROVEN]** for periodic schedules; **[SUPERSEDED]** for arbitrary deltas	Theorem 4.2 here; Theorem 6.1; FL Lemma 12
Sandwich packet layout and the other two layouts	BioChain AI 2026e, 2026f, 2026g	**[SUPERSEDED]**	By the Einstein Tile; Definition 8.1; Section 4.6 here

Claim	Source	Status	Basis / Note
Lossless TQF pruning	BioChain AI 2026e, Stage 3	**[SUPERSEDED]**; zero second difference **[PROVEN]**	Proposition 8.3
24 streams collapse to 9 solitons	BioChain AI 2026e, Stage 2	**[OPEN]**	No rule fixes nine; "3 of 9" is FL Proposition 8.2″; Problem 5 here
Zeta energy filter	BioChain AI 2026f §2.1	**[HEURISTIC]**	Well defined once restated with a modulus; Section 4.7 here
S^3 parallelizable; scalar key	BioChain AI 2026h	**[ESTABLISHED]**	Section 4.1; Proposition 7.7
Φ as a truth score	BioChain AI 2026h	**[HEURISTIC]**; invariance **[PROVEN]**	Frame-invariant, not sign-invariant; Proposition 9.1; Section 4.8 here
Cubic reconstruction from triple-quad measurements	BioChain AI 2026i	**[HEURISTIC]**	An inverse problem; uniqueness not shown; Problem 7 here
South Halo as an integrity check	BioChain AI 2026f §3.1	**[SUPERSEDED]**	A cache filter (FL Lemma 15); Section 4.7 here
Gibbs state positive on every node	ed. 1; FL	**[PROVEN]**	Proposition 2.2; FL Lemma 14
Prime traps give the harmonic measure	ed. 1; FL	**[PROVEN]**	Proposition 2.3; FL Lemma 14
Primes ≡ 7 mod 8 are never 3D quadrances	ed. 1	**[ESTABLISHED]**	Theorem 4.2
No collinear triple of prime quadrances	ed. 1; FL	**[PROVEN]**	Theorem 8.4; FL Lemma 5
The Engine's tier rotation is a Hurwitz unit	ed. 1; FL	**[PROVEN]**	Proposition 10.1; FL Lemma 4
Killing concentrates but never selects	ed. 2	**[PROVEN]**	Proposition 2.4
The sieve on the infinite Hurwitz lattice	FL; sharpness ed. 2	**[PROVEN]**	Proposition 2.5
Drift without a potential does not sieve	FL	**[PROVEN]**	Proposition 2.6
Drift without a potential produces entropy at $\Sigma(1-\rho_k)w_k \ln(1/\rho_k)/W$	release; Hoard 2026d	**[PROVEN]**	Proposition 2.6; Hoard 2026d, Proposition 4.8
A potential carries no stationary current; a potential-free drift selects nothing	release; Hoard 2026d	**[PROVEN]**	Remark (Kolmogorov's Criterion), Section 2.3
Certified traps are exact	ed. 2	**[PROVEN]**	Proposition 3.1; FL Lemma 14 Part 4
The wheel's bias is 3.75	ed. 2	**[PROVEN]**	Proposition 3.2
$\mathbb{Z}/30$ is the largest primorial wheel fitting a 3-bit field	release; Hoard 2026d	**[PROVEN]**	Proposition 3.3; Hoard 2026d, Theorem 4.9

Claim	Source	Status	Basis / Note
Primes of the box $[-2, 2]^4$ are 2, 3, 5, 7 and 13	FL	**[PROVEN]**	Section 4.1; FL Lemma 5
Canonical associate for addresses; sign normalization for rotations	ed. 2; FL	**[PROVEN]**, **[DEFINITION]**	Proposition 4.3
Torus-preserving delta	ed. 2	**[PROVEN]**	Proposition 5.1
No lattice point on the golden torus	ed. 2	**[PROVEN]**	Theorem 5.2
Fibonacci-torus points; Lucas primes ≡ 7 mod 8	ed. 2	**[PROVEN]**	Theorem 5.3
Two-gap golden circle at Fibonacci N	FL	**[ESTABLISHED]**	Section 5.3; FL Lemma 11
Exponents 12, 24, 60 close periodic schedules	ed. 2	**[PROVEN]**	Theorem 4.2 here
Exact deltas close within 12 repetitions or never	FL; corollary ed. 2	**[PROVEN]**	Theorem 6.1; FL Lemma 13
Exact arithmetic has no drift	ed. 2; FL	**[PROVEN]**	Proposition 6.2; FL Lemma 13
Norm-5 words are free with unique addresses	ed. 2	**[ESTABLISHED]**, **[COMPUTED]**	Theorem 7.2
Generation by shift and add	ed. 2	**[PROVEN]**	Proposition 7.3
Ramanujan meeting graphs mix in logarithmic time	ed. 2	**[ESTABLISHED]**, **[COMPUTED]**	Theorem 7.4; Proposition 7.5
The {4,3,5} honeycomb as a second hyperbolic cube structure	FL	**[ESTABLISHED]**, **[PROVEN]**	Section 7.4; FL Lemma 8 and Theorem 7.4
$C_3 \times C_2$ is the abelian shadow of PSL(2, $\mathbb{Z}$); SL(2, 3) = 2T; SL(2, 5) = 2I	ed. 2	**[ESTABLISHED]**, **[COMPUTED]**	Proposition 7.6
Scalar zero is the ideal boundary; the ghost map is a rational isometry	ed. 2; FL	**[PROVEN]**	Propositions 7.7 and 7.8; FL Lemma 8
CRT addresses with closed-form inverses	FL	**[PROVEN]**	Section 7.7; FL Lemma 9
The Einstein Tile as the packet	FL	**[DEFINITION]**	Section 8.1
The cubit as the unit of the scale register	release; Hoard 2026d	**[DEFINITION]**; voxel facts **[PROVEN]**	Section 8.1; Hoard 2026d, Proposition 3.1
Packet arithmetic is exact in 64 bits	ed. 2; FL	**[PROVEN]**	Proposition 8.2; FL Lemma 5
Precision horizon $Q \approx 2.3 \times 10^7$ for cubit-anchored lattices	release; Hoard 2026d	**[HEURISTIC / COMPUTED]**	Section 8.2; Hoard 2026d, §3.5
Double precision misjudges collinearity at the source-note horizon	release; Hoard 2026d	**[COMPUTED]**	Section 8.2; Hoard 2026d, Remark 4.3

Claim	Source	Status	Basis / Note
TQF pruning removes exactly the zero-second-difference nodes	ed. 2	**[PROVEN]**	Proposition 8.3
Joint 0.5 s window for caesium, 40 Hz and 432 Hz	Hoard 2026d; FL	**[PROVEN]**	Section 9.2; Hoard 2026d, §4.1; FL Proposition 11.4
Frame-free invariants; corrected coherence Φ_M	ed. 2	**[PROVEN]**	Proposition 9.1
Cubit-scaled lengths give medium-free invariants	release; Hoard 2026d	**[PROVEN]**	Section 9.3; Hoard 2026d, Lemma 3.2
The Engine's tier rotations generate the certification group O	release; Hoard 2026d	**[PROVEN]**	Section 9.4; Hoard 2026d, Appendix E
Merkle roots certify; groups of four survive one loss	FL	**[PROVEN]**	Section 9.6; FL Lemmas 7 and 15
The Engine's 60 and 30 in the hyperbolic constructions	ed. 2	**[CORRESPONDENCE]**	Remarks in Sections 6 and 10
The lattice standard as an AI alignment and governance layer	ed. 2	**[HEURISTIC]**	Section 9; Problem 8 here

3 Audit of the Source Note

The source note *The Fokker-Planck Prime Sieve in Quaternionic Lattices* is a three-page manuscript. Table 4 lists every point at which Hoard (2026e) departs from it. Items 7, 11 and 12 overturn a stated claim; the others correct a formulation, a number or a missing step.

Table 4: Errata and audit of the source note.

#	Location in the note	The note states	Finding	Resolution in Hoard 2026e
1	Abstract, §1	A 4 × 4 × 4 lattice for a four-component state q	A 3D grid cannot hold 4D states and misses primes ≡ 7 mod 8	4D Hurwitz lattice (4.1)
2	§1	Partial derivatives in q on a discrete lattice	Not defined on nodes	Master equation (2.1)
3	§1	$D_1 = -\nabla V(Q)$ governed by TQF trajectories	V depends only on Q, so its gradient is radial	Stated (2.1)
4	§2	ρ = Σ c_p(t) Φ_p e^(−λ_p t)	Time dependence counted twice	Constant coefficients (2.1)
5	§2	A prime-indexed basis	Cannot represent a uniform start	Eigen-expansion of the generator (2.1)
6	§2	Composite modes have high λ	Not derived	Absorbing traps (2.2)
7	§3	Inert primes exist but are rare	No prime is inert in the Hurwitz order	Proposition 4.1
8	§3	"The Prime Conflict in 3D"	The argument given is about four squares	Theorem 4.2

#	Location in the note	The note states	Finding	Resolution in Hoard 2026e
9	§3	Stability needs a primitive path vector	Automatic for prime norm	Section 4.1
10	§4	Example Q = 2, 3, 5 on a TQF line	A_R = 24: a right triangle, not collinear	Below
11	§4	Prime traps where TQF lines meet prime nodes	No collinear triple has three prime quadrances	Theorem 8.4
12	§5	Output supported only on prime nodes	False for any positive diffusion or killing	Propositions 2.2 to 2.4; certified traps (3.1)
13	§6	About 60 % of a 64-bit word	40 bits are 62.5 %	Confirmed (8.2)
14	§6	A 64-bit lattice is vastly sufficient	True at packet scale and for cubit-anchored lattices; the TQF needs 85 bits at the horizon	Proposition 8.2; Table 13
15	§6	Text ends "well within the 'sweet"	Conclusion of §6 missing	Supply the sentence

The example triple. The note's example Q = 2, 3, 5 gives $A_R = (2 + 3 + 5)^2 - 2(4 + 9 + 25) = 24 = 16A^2$: since 2 + 3 = 5 it is a right triangle, not a collinear one. The same confusion of the right-angle condition $Q_1 + Q_2 = Q_3$ with collinearity recurs in the decoding framework (Hoard 2026e, Proposition 8.3(3); Table 5 here, item 12). The Fifteen Lemmas realize the triangle in $[-2, 2]^3$ with A = (0, 0, 0), B = (1, 1, 1) and C = (1, 0, 2). If the Q_i instead denote the norms of three nodes on one lattice line $q(t) = q_0 + td$, the norm along the line is the quadratic $N(q_0) + 2t\langle q_0, d\rangle + t^2N(d)$, and whether it is prime infinitely often is Problem 2 here.

The precision horizon. The note's resolution rule $\delta\theta \approx 1/\sqrt{Q}$ is a heuristic, and its consequences are computed: $Q \geq 10^{12}$ for six decimal places, coordinates near 10^6, and 40 of 64 bits. The release adds a physical ceiling. For lattices anchored to the realized cubit the same rule gives $Q \approx 2.3 \times 10^7$, 44,000 times lower, because the cubit is realized only to a relative standard uncertainty of 2.09×10^{-4} (Hoard 2026e, Section 8.2; Hoard 2026d, §3.5). Both horizons are tagged **[HEURISTIC / COMPUTED]**; Problem 10 here tests the rule itself.

4 Corrections of the Companion Papers

The sieve uses nine companion papers of the SHD-CCP programme. Where one of them states a claim that the release replaces, this section gives the claim, the reason, and the replacement; Table 5 collects the items. A correction that rests on a theorem is proved here or in Hoard (2026e).

Status: [PROVEN], [COMPUTED] and [SUPERSEDED]. Sources: BioChain AI (2026c), (2026d), (2026e), (2026f), (2026g) and (2026h); Fifteen Lemmas, Lemmas 8, 11, 12 and 15; Hoard (2026d), Proposition 4.6 and Remark 4.7.

4.1 The Golden Clifford Torus

Correction (One-Sided Delta). BioChain AI (2026c), §3.1, applies $q \mapsto e^{i\Delta\theta_1} e^{j\Delta\theta_2} q$. That product is a rotation of S^3, but it does not preserve the torus: in 1,000 random trials the

result left the torus every time (Hoard 2026e, Appendix A, section L). The two-sided form e^{iα} q e^{iβ} of Hoard (2026e), Proposition 5.1, is exact in all 1,000.

Correction (Ergodic Covering). BioChain AI (2026c), §2.2, derives a dense covering from the radius ratio $r_1/r_2 = \varphi$. Whether a stream winds densely over the torus depends on the ratio of its angular rates, not on the ratio of the radii: a stream $\theta_1 = \omega_1 t$, $\theta_2 = \omega_2 t$ is dense exactly when ω_1/ω_2 is irrational, and golden radii do not force it (Fifteen Lemmas, Lemma 11 Parts 2 and 5) **[PROVEN]**. Choosing $\omega_1/\omega_2 = \varphi$ realizes the source's intent.

Correction (Golden Solitons). BioChain AI (2026c), §4.1, places "stable vertices" where streams collide "strictly where the generalized quadrance equals a rational prime". No point of $\mathbb{Q}^4$ other than 0 lies on a Clifford torus with radius ratio φ (Hoard 2026e, Theorem 5.2), so these cannot be lattice nodes on the torus **[SUPERSEDED]**. The exact nodes nearest the torus lie on its Fibonacci approximants, with prime norms 3, 7, 47 and 2207 (Hoard 2026e, Theorem 5.3).

Correction (The Strassen Step). BioChain AI (2026c), §5, states that a structured quaternion step needs "only 16 core multiplications" by a Strassen-like scheme. Sixteen real multiplications is the cost of the direct Hamilton product; no Strassen reduction is derived **[SUPERSEDED]**. Steps by the norm-5 generators need no multiplier at all (Hoard 2026e, Proposition 7.3).

4.2 Quaternion Matrices and Inverses

Correction (Quaternion Matrices). BioChain AI (2026d), §2.2, calls the 4 × 4 matrix of q = w + xi + yj + zk a real skew-symmetric matrix. It is the matrix L(q) of left multiplication by q, which satisfies $L(q)^T L(q) = N(q)I$ and $L(q) + L(q)^T = 2wI$: a scaled orthogonal matrix with $\det L(q) = N(q)^2$, skew-symmetric only when w = 0 (Fifteen Lemmas, Lemma 8, Step 2) **[SUPERSEDED]**.

Correction (Inverses). BioChain AI (2026d), §2.1, states that every generated Hurwitz path has an inverse. The inverse $\bar{q}/N(q)$ of a non-unit is not a Hurwitz integer, so the claim holds over $\mathbb{Q}$ only. Exact inversion of a norm-5 word is instead the reversed word of conjugates, divided by the scalar 5^L (Hoard 2026e, Definition 7.1).

4.3 Trefoil Folding

The symmetry-folding paper routes three data streams along the trefoil x = sin t + 2 sin 2t, y = cos t − 2 cos 2t, z = −sin 3t, and stores one third of it (BioChain AI 2026d, §4).

> **Proposition 4.1 (Trefoil Symmetry).** **[PROVEN]** Fifteen Lemmas, Lemma 8 Step 6; confirms BioChain AI (2026d), §4.2.
>
> *The curve is carried to itself by the rotation of 120° about the z-axis, with $t \mapsto t - 2\pi/3$, and by the half-turn about the y-axis, with $t \mapsto -t$. These generate the dihedral group D_3 of order 6. With the C_3 axis along (1, 1, 1) and a half-turn axis along (1, −1, 0), its six elements are integer signed-permutation matrices in the cube group O.*
>
> *Proof. Write w(t) = x(t) + iy(t) = ie^{−it} − 2ie^{2it}. Then $w(t + 2\pi/3) = e^{-2\pi i/3} w(t)$ and $z(t + 2\pi/3) = z(t)$, a rotation by 120°; and (x, y, z)(−t) = (−x, y, −z)(t), the half-turn about y. Conjugating by the orthonormal frame whose second axis is (1, −1, 0)/√2 and third axis (1, 1, 1)/√3 turns the two generators into the*

integer matrices with rows (0, 1, 0), (0, 0, 1), (1, 0, 0) and (0, −1, 0), (−1, 0, 0), (0, 0, −1). ■

Hoard (2026e), Appendix A, section N, confirms both identities to 4×10^{-15}, and section X that the two integer matrices generate six rotations.

Correction (What Folding Saves). With the full D_3, a fundamental arc is one sixth of the curve, not one third: the arc $t \in [0, \pi/3]$ and its six integer images rebuild the whole knot (Fifteen Lemmas, Lemma 8 Step 6), so the curve tables shrink by five sixths. Data written along the knot is a different matter: folding it is lossless only if the data itself is D_3-symmetric, because a map from six segments to one is injective only on symmetric inputs. The source's 66.6 % memory saving holds for the tables, not for the data stream **[PROVEN]**.

4.4 The 720-Step Cycle

The source restricts evolution to a 720-step state machine and states that at step 720 "the product of all quaternion deltas, trefoil symmetries, and hyperbolic projections mathematically guarantees a return to the Identity Matrix" (BioChain AI 2026d, §5). The guarantee holds for a definite class of schedules, and in exact arithmetic that class is small.

Theorem 4.2 (Finite Delta Groups Close). **[PROVEN]** Restricts and proves BioChain AI (2026d), §5.

The binary tetrahedral, octahedral and icosahedral groups 2T, 2O and 2I have exponents 12, 24 and 60, each dividing 720. Hence if a schedule repeats one fixed word w in the deltas of any of these groups, $w^{720} = 1$, and already $w^{12} = 1$, $w^{24} = 1$ or $w^{60} = 1$ respectively.

Proof. The element orders are 1, 2, 3, 4 and 6 in 2T; 1, 2, 3, 4, 6 and 8 in 2O; and 1, 2, 3, 4, 5, 6 and 10 in 2I (Hoard 2026e, Appendix A, section J). Their least common multiples are 12, 24 and 60. The word w is a group element, so its power by the exponent is 1. ■

By Hoard (2026e), Theorem 6.1, only the elements of 2T and those of orders 1, 2, 3, 4 and 6 in 2O and 2I can be realized in exact rational arithmetic, so an exact periodic schedule that returns at all returns within 12 repetitions.

Remark (Arbitrary Deltas Do Not Close). **[COMPUTED]** When the 720 deltas are drawn independently from the 24 Hurwitz units, the product equals 1 in 92 of 2,000 runs (4.6 %), close to the 1/24 expected of a uniformly random element. The source's guarantee for arbitrary deltas is **[SUPERSEDED]**.

Remark (The Cyclic Holonomy Is Vacuous). **[PROVEN]** The cubit installment proves the identity behind the guarantee: with the cyclic convention $q_{-1} := q_{719}$ and increments $\Delta q_k = q_k\, q_{(k-1)}^{-1}$, the product of all 720 increments equals 1 for every sequence of states, by telescoping (Hoard 2026d, Proposition 4.6(1)). A closure test built from stored states therefore cannot fail and tests nothing. The Fifteen Lemmas reach the same verdict from the holonomy side: under no boundary convention does the 720-frame cycle force the state back to the identity or remove drift, and only the vacuous cyclic convention makes the holonomy identically 1 (Lemma 12). A genuine test needs increments generated independently of the states, for example by integrating an angular velocity, a stated product order, and a measure

such as the rotation angle 2 arccos|Re H| (Hoard 2026d, Remark 4.7); the open convention $H = q_0 q_{719}^{-1}$ is such a test. Drift itself is removed by exact arithmetic (Hoard 2026e, Proposition 6.2), not by the cycle.

4.5 The Hyperbolic Projection

The source maps quaternion rotations into SL(2, $\mathbb{C}$), the double cover of the Lorentz group $SO^+(3, 1)$, so that branching unfolds in hyperbolic 3-space (BioChain AI 2026d, §3) **[ESTABLISHED]**. The precise statement is an inclusion: $SU(2) \subset SL(2, \mathbb{C})$ is the stabilizer of the base point of hyperbolic space, and there is no homomorphism $SO(3) \to SL(2, \mathbb{C})$ splitting the cover (Fifteen Lemmas, Lemma 8, Proposition 6.3). The projection matrices are exact only if their entries lie in a ring such as the Gaussian integers $\mathbb{Z}[i]$; with real entries drift returns. The release takes two exact routes instead: integer quaternions acting on a tree, and the rational ghost map (Hoard 2026e, Sections 7.1 and 7.6).

4.6 Packet Layouts

Three companion papers described the 64 bits of the packet, and they disagreed: a sandwich of an 8-bit North Halo, 16 scaling bits, a 32-bit quaternion core and an 8-bit South Halo (BioChain AI 2026e); four 16-bit phases X, Y, Z, W (BioChain AI 2026f); and 16 fibres × 4 phases (BioChain AI 2026g). The release adopts the Einstein Tile (Hoard 2026e, Definition 8.1), onto which the earlier layouts map: the sandwich's 32-bit core is the core word, its halos are the header and gate bytes, and its 16 scaling bits are the scale register. As field maps they are **[SUPERSEDED]**; the 16-fibre × 4-phase multiplex of BioChain AI (2026g) survives as an addressing of the same 64 bits.

4.7 Claims About the Protocol That Do Not Survive

- **Lossless pruning (BioChain AI 2026e).** **[SUPERSEDED]** Lossless only in the sense of Hoard (2026e), Proposition 8.3(2).
- **The South Halo as an integrity check (BioChain AI 2026f, §3.1).** **[SUPERSEDED]** An 8-bit header splits the 2^{56} payloads into 256 classes of 2^{48} each on average, so it is a fast pre-filter and never an integrity check (Fifteen Lemmas, Lemma 15 Part 2). Integrity comes from a Merkle root (Hoard 2026e, Section 9.6).
- **The zeta filter (BioChain AI 2026f, §2.1).** **[HEURISTIC]** The energy check $E > \zeta(\tfrac{1}{2} + it_7)$ compares a real number with a complex one. It becomes a well-defined filter once restated with a modulus, $E > |\zeta(\tfrac{1}{2} + it_7)|$, and stays a heuristic.
- **Twenty-four streams collapsing to nine solitons (BioChain AI 2026e, Stage 2).** **[OPEN]** No rule is given that fixes nine; Problem 5 here.

4.8 The Coherence Score

The voxel-alignment review presents $\Phi = (1/N)\|\Sigma\, \hat{q}_i\|$ as a score that "mathematically guarantees the integrity" of a packet (BioChain AI 2026h). Φ is invariant under frame changes but not under $q_i \mapsto -q_i$, which represents the same rotation, and it measures agreement among packets, so a coherent error scores as high as a coherent truth. Hoard (2026e), Proposition 9.1, gives the sign-invariant form Φ_M, and Section 9.6 there keeps it as an advisory Helix A score **[HEURISTIC]**.

4.9 Errata of the Companion Papers

Table 5: Errata and audit of the companion papers.

#	Paper and location	The paper states	Finding	Resolution
1	Hoard 2026a, §III.2	Angular velocity "exponentially accelerates" toward the core	ω_n is proportional to W_n = 16 − 2n: linear, doubling from tier 4 to tier 0	Wording only; no effect on the sieve
2	BioChain AI 2026b, Definition 3.1	Golden shear cφ/N in the metric	Not invariant under c → c + N	Superseded in BioChain AI 2026a, §3.2
3	BioChain AI 2026c, §2.2	Dense covering because $r_1/r_2 = \varphi$	Density depends on the ratio of angular rates	Section 4.1 here
4	BioChain AI 2026c, §3.1	Δq = e^{iΔθ₁} e^{jΔθ₂}, applied on the left	Leaves the torus in 1,000 of 1,000 trials	Two-sided delta, Proposition 5.1
5	BioChain AI 2026c, §4.1	Solitons form strictly at prime quadrances on the torus	No lattice point lies on the golden torus	Theorems 5.2 and 5.3
6	BioChain AI 2026c, §5	16 multiplications by a Strassen-like scheme	16 is the direct Hamilton product	Proposition 7.3
7	BioChain AI 2026d, §2.1	Every generated Hurwitz path has an inverse	The inverse $\bar{q}/N(q)$ of a non-unit is not a Hurwitz integer	Section 4.2 here
8	BioChain AI 2026d, §2.2	Quaternion matrices are skew-symmetric	$L(q)^T L(q) = N(q)I$; skew only for w = 0	Section 4.2 here
9	BioChain AI 2026d, §4.2	D_3 folding cuts memory by 66.6 %	True for the curve tables, not for data	Section 4.3 here
10	BioChain AI 2026d, §5	T_n ≡ T_n (mod 720); identity guaranteed at step 720	The congruence is a tautology; the cyclic holonomy is identically 1; arbitrary deltas close in 4.6 % of runs	Theorem 4.2 and Section 4.4 here
11	BioChain AI 2026e, Stage 3	TQF pruning on scan lines with no data loss	Every scan-line triple is collinear	Proposition 8.3
12	BioChain AI 2026e, Stage 3 figure	$Q_1 + Q_2 = Q_3$ marks a linear node	That is the right-angle condition	Proposition 8.3(3)
13	BioChain AI 2026e, 2026f, 2026g	Three different maps of the 64 bits	Mutually inconsistent	Einstein Tile, Definition 8.1; Section 4.6 here
14	BioChain AI 2026f, §2.1	$E > \zeta(\tfrac{1}{2} + it_7)$	Compares a real with a complex number	Section 4.7 here
15	BioChain AI 2026f, §2.2	TQF collinearity allows projection to 2D "with zero loss of fidelity"	Collinearity alone does not fix a point's position on its line	Proposition 8.3(2)
16	BioChain AI 2026h	Φ "mathematically guarantees the integrity" of a packet	Φ is not sign-invariant and measures agreement, not truth	Proposition 9.1; Section 4.8 here
17	BioChain AI 2026d, §5	The 720-frame cycle removes floating-point drift	No boundary convention makes the cycle remove drift (Fifteen Lemmas, Lemma 12; Hoard 2026d, Proposition 4.6)	Theorem 6.1; Proposition 6.2; Section 4.4 here

#	Paper and location	The paper states	Finding	Resolution
18	BioChain AI 2026f, §3.1	The South Halo carries a parity checksum for neighbour verification	An 8-bit header has 2^{48} payloads per value on average, and a linear checksum can be steered	Section 4.7 here; integrity by Merkle root, Section 9.6

5 Open Problems

Ten problems remain open. Each is stated in the manner of a registered report, following the cubit installment (Hoard 2026d, Section 8): the evidence so far, a test with its decision rule fixed before any new computation or measurement, and whether the release depends on the answer. It depends on none of them. A computation can refute a conjecture about infinitely many objects but never prove it, and the decision rules say so.

Table 6: Open problems, their tests and their bearing on the release.

#	Problem	Status	Test	Release depends on it?
1	Infinitely many prime Lucas numbers L_{2^k}	**[OPEN]**	Primality of L_{2^k} for further k	No
2	A lattice line with infinitely many prime-norm nodes	**[OPEN]**	Counts against the Bateman–Horn prediction	No
3	Oracle bias above 1 for factoring	**[CONJECTURAL]**	Preregistered semiprime benchmark	No
4	Trefoil knot invariants and prime quadrances	**[OPEN]**	Requires a stated mechanism first	No
5	A rule that fixes nine solitons	**[OPEN]**	Requires a stated rule first	No
6	Positive flux for drifts that vary from cell to cell	**[OPEN]**	Exhaustive search on small tori	No
7	Unique cubic reconstruction from triple-quad measurements	**[OPEN]**	Search for distinct cubes with equal measurements	No
8	The alignment handshake between real AI systems	**[HEURISTIC]**	Protocol H	No; it tests the application
9	A prime potential defined without primality	**[OPEN]**	The falsifiable test of the first edition	No
10	The resolution rule $\delta\theta \approx 1/\sqrt{Q}$	**[HEURISTIC]**	Covering radius of lattice directions	No; it fixes the horizons of Section 8.2

Problem 1 (Lucas Primes on the Fibonacci Tori). [OPEN] Raised by Hoard (2026e), Theorem 5.3.

Are infinitely many of the Lucas numbers L_{2^k} prime?

Evidence. $L_2 = 3$, $L_4 = 7$, $L_8 = 47$ and $L_{16} = 2207$ are prime; $L_{32} = 4{,}870{,}847 = 1087 \times 4481$ is composite, and among even $n \le 800$ no other L_n is prime (Hoard 2026e, Appendix A, section M). The heuristic that a number near x is prime with probability about 1/ln x gives the sum over k of a multiple of 2^{−k}, which converges, so finitely many are expected, as for the Fermat primes.

Test and decision rule. Certify L_{2^k} for k = 10 to 14 by a proven primality test or a factor. A prime refutes nothing; the problem stays open either way.

Dependence. None. The release uses only the four known primes.

Problem 2 (Prime Norms Along a Lattice Line). **[OPEN]** Raised by the first edition; Hoard (2026e), Section 8.3.

Does a lattice line $q(t) = q_0 + td$, on which $N(q(t)) = N(q_0) + 2t\langle q_0, d\rangle + t^2N(d)$ is irreducible with no fixed prime divisor, meet infinitely many nodes of prime norm?

Evidence. This is a case of Bunyakovsky's conjecture, open for every quadratic polynomial.

Test and decision rule. For 100 random primitive lines with coordinates below 100, count the prime norms for $0 \le t \le 10^7$ and compare with the Bateman–Horn prediction (Bateman and Horn 1962). A line whose count falls below one tenth of the prediction refutes the heuristic constant for that line; agreement within 10 % on all lines is recorded as support, not proof.

Dependence. None. Theorem 8.4 already limits prime traps on a line to two.

Problem 3 (Oracle Bias for Factoring). **[CONJECTURAL]** Raised by BioChain AI (2026a), §5.

Does a geometric oracle raise the probability that a candidate yields a factor above its uniform value?

Evidence. For the prime task the bias is proven: 3.75 for the 30-node wheel (Hoard 2026e, Propositions 3.2 and 3.3). For factoring no bias has been measured.

Test and decision rule. On a preregistered set of 10,000 semiprimes of fixed size, compare the success rate of the oracle-ranked search with uniform ranking at equal cost. The conjecture is supported if the ratio's 95 % confidence interval lies above 1 and refuted if it lies at or below 1.

Dependence. None. The release makes no claim about factoring.

Problem 4 (Knot Invariants and Prime Quadrances). **[OPEN]** Raised by BioChain AI (2026d), §8, horizon 1.

Do knot invariants of the trefoil predict prime quadrances?

Evidence. None. The trefoil's D_3 symmetry is proven (Proposition 4.1 here), but no map from a knot invariant to a quadrance has been proposed.

Test and decision rule. Before any test, a mechanism must name the invariant, the quadrance it predicts and the map between them. Without one the problem should be retired.

Dependence. None.

Problem 5 (Nine Solitons). **[OPEN]** Raised by BioChain AI (2026e), Stage 2.

What rule collapses 24 streams to 9 solitons?

Evidence. None given. The related rule "3 of 9 streams pass to the cube" has an exact reading: exactly three of the nine C_9 rays end at cube vertices (Fifteen Lemmas, Lemma 8, Proposition 8.2″).

Test and decision rule. As for Problem 4, a stated rule must come first; it is then checked exhaustively over the 24 streams.

Dependence. None.

Problem 6 (Flux Under Cell-Varying Drift). [OPEN] Raised by Fifteen Lemmas, companion Problem P1; Hoard (2026e), Proposition 2.6.

On $(\mathbb{Z}/4)^3$, let every forward step $+e_k$ carry more probability than the reverse step from the neighbouring cell, with rates that vary from cell to cell. Is the net stationary flux through every cross-section perpendicular to e_k positive?

Evidence. Edge by edge the statement fails, while the cross-sectional flux was positive in every trial. By Kolmogorov's criterion a current exists exactly when the products of rates around some cycle differ (Hoard 2026d, Proposition 4.8(3)), so the question is whether those cycle imbalances always add up to a positive flux.

Test and decision rule. Draw 10^6 random drifts satisfying the hypothesis, compute the stationary state exactly in rational arithmetic, and evaluate every cross-sectional flux. One negative flux refutes the conjecture; none is recorded as support, and a proof is then sought by the tree theorem.

Dependence. None. Proposition 2.6 covers the translation-invariant case the sieve uses.

Problem 7 (Cubic Reconstruction). [OPEN] Raised by BioChain AI (2026i).

Do the triple-quad measurements of a 3-folded packet determine the cube uniquely, up to its symmetries?

Evidence. The paper states the inverse problem but shows no uniqueness.

Test and decision rule. Enumerate the integer cubes of edge at most 4 and their measurement vectors. Two inequivalent cubes with equal vectors refute uniqueness; none, at that size, is recorded as support.

Dependence. None. The release uses scale-normalized quads only as signatures, and Section 9.4 there resolves ambiguous signatures by certification.

Problem 8 (The Alignment Handshake in Practice). [HEURISTIC] Raised by Hoard (2026e), Section 9.

Does the alignment handshake let two independently built AI systems exchange states exactly and certify their relative frame?

Evidence. In simulation, signature matching succeeded in 499 of 500 trials and exact certification in all 500 (Hoard 2026e, Section 9.4). No test with real systems has been run.

Test and decision rule (Protocol H). Two systems with independently trained encoders each map the same 1,000 preregistered objects into the lattice and run the handshake of Hoard (2026e), Figure 1. Primary outcome: the fraction of sessions that end in an exact certificate. Secondary outcomes: the number of rounds, and the fraction of certified anchor pairs that a blinded human panel judges to denote the same object. Every certificate is exact by construction, so a single certificate that fails independent re-verification refutes the implementation. The protocol is supported if at least 95 % of sessions certify and the panel agreement exceeds the agreement of randomly paired anchors by a margin fixed in advance. Semantic agreement is reported separately, because no certificate can establish it.

Dependence. None for the mathematics; the protocol tests the application.

Problem 9 (A Prime Potential Without Primality). **[OPEN]** Raised by the first edition.

Is there a potential on the lattice, defined without a primality test or a factorization of Q, whose minima are exactly the prime-quadrance nodes?

Evidence. None. Under the Dual-Helix principle the sieve does not need one (Hoard 2026e, Section 3).

Test and decision rule. The first edition's falsifiable test: a proposed rule is refuted if its minima miss a prime quadrance or include a composite one, checked for all $Q \le 10^6$.

Dependence. None.

Problem 10 (The Resolution Rule). **[HEURISTIC]** Raised by the source note, §6; Hoard (2026e), Section 8.2.

Is $\delta\theta \approx 1/\sqrt{Q}$ the right angular resolution of the lattice directions of norm at most Q?

Evidence. The rule sets both precision horizons of the release, $Q \approx 10^{12}$ for six decimal places and $Q \approx 2.3 \times 10^7$ for cubit-anchored lattices. In three dimensions the lattice directions of norm at most Q are spaced about Q^{−3/4} apart in angle on average, while the largest gaps, next to directions of small norm, are of order Q^{−1/2}, so the rule matches the worst case rather than the average.

Test and decision rule. Compute the covering radius of the primitive directions of norm at most Q on S^2 and S^3 for Q up to 10^6. The rule is confirmed if $\sqrt{Q}$ times the covering radius stays within a fixed factor of 1 across that range, and replaced by the fitted law otherwise; the horizons of Section 8.2 are then recomputed from it.

Dependence. Only the horizons, which are tagged **[HEURISTIC / COMPUTED]**; no proof uses them.

6 Verification

The script verify_fp_sieve.py of Hoard (2026e), Appendix A, reproduces every number in this companion. Table 7 lists the sections that check it.

Table 7: Sections of the verification script that check this companion.

Section	What it checks here
A, H	Arithmetic and widths of the source note, including A_R(2, 3, 5) = 24 (Section 3)
B	No collinear prime triple, which settles item 11 of Table 4 (Section 3)
J	Theorem 4.2; random 720-step products close in 92 of 2,000 runs (Section 4.4)
L	The one-sided delta leaves the torus in 1,000 of 1,000 trials (Section 4.1)
M	Lucas numbers L_n for even n ≤ 800 (Problem 1)
N, X	Proposition 4.1: trefoil symmetries and the six integer rotations of D_3 (Section 4.3)
O	The right-angle reading $Q_1 + Q_2 = Q_3$; the identities of L(q) (Sections 3 and 4.2)
R	Statistics of the second edition's byte core, kept for comparison (Section 1.3)
S	Φ is frame-invariant but not sign-invariant (Section 4.8)
Y	1,712 of 1,877 primes reached by the Form 1 core (Section 1.3)
Z	The cubit results that the release adds (Section 1.1)

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